

Restaurant Ratings in Mexico: Case Study

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Contents

1	Case Study	2
2	Dataset Description	2
2.1	Consumers	2
2.2	Consumer Preferences	2
2.3	Ratings	3
2.4	Restaurants	3
2.5	Restaurant Cuisines	3
3	ER Diagram	4
4	Data Cleaning	4
4.1	Steps to Import Data as a Folder	4
4.2	Calculated Fields	4
4.2.1	Age Group	4
4.2.2	Service Rating Category	5
4.2.3	Overall Rating Category	5
4.2.4	Food Rating Category	5
5	Data Analysis	5
5.1	Local Insights	5
5.2	Dining Insights	5
5.3	Hospitality Insights	6
5.4	Behavior Insights	6
5.5	Review Insights	6
5.5.1	Top Restaurants by Food Rating	6
5.5.2	Top Restaurants by Service Rating	6
5.5.3	Top Restaurants by Overall Rating	6

1 Case Study

Restaurant ratings in Mexico by real consumers from 2012, including additional information about each restaurant and their cuisines, and each consumer and their preferences.

2 Dataset Description

Our dataset consists of the following observations:

2.1 Consumers

- **Consumer_ID** – Unique identifier for each consumer
- **City** – City where the consumer lives
- **State** – State where the consumer lives
- **Country** – Country where the consumer lives
- **Latitude** – Latitude where the consumer lives
- **Longitude** – Longitude where the consumer lives
- **Smoker** – Whether the consumer smokes or not
- **Drink_Level** – Whether the consumer is an abstemious, casual, or social drinker
- **Transportation_Method** – Whether the consumer transports on foot, by public transport, or by car
- **Marital_Status** – The consumer’s marital status (single or married)
- **Children** – Whether the consumer has dependent/independent children or kids
- **Age** – The consumer’s age
- **Occupation** – The consumer’s occupation (student, employed, or unemployed)
- **Budget** – The consumer’s budget (low, medium, high)

2.2 Consumer Preferences

- **Consumer_ID** – Unique identifier for each consumer
- **Preferred_Cuisine** – Types of food the consumer prefers

2.3 Ratings

- **Consumer_ID** – Unique identifier for each consumer
- **Restaurant_ID** – Unique identifier for each restaurant
- **Overall_Rating** – Overall rating by the consumer (0 = Unsatisfactory, 1 = Satisfactory, 2 = Highly Satisfactory)
- **Food_Rating** – Food rating by the consumer (0 = Unsatisfactory, 1 = Satisfactory, 2 = Highly Satisfactory)
- **Service_Rating** – Service rating by the consumer (0 = Unsatisfactory, 1 = Satisfactory, 2 = Highly Satisfactory)

2.4 Restaurants

- **Restaurant_ID** – Unique identifier for each restaurant
- **Name** – Restaurant name
- **City** – Restaurant city
- **State** – Restaurant state
- **Country** – Restaurant country
- **Zip_Code** – Restaurant ZIP code
- **Latitude** – Restaurant latitude
- **Longitude** – Restaurant longitude
- **Alcohol_Service** – Whether the restaurant serves no alcohol, wine and beer, or a full bar
- **Smoking_Allowed** – Whether smoking is allowed
- **Price** – Restaurant price level (low, medium, high)
- **Franchise** – Whether the restaurant is a franchise
- **Area** – Whether the restaurant is in an open or closed area
- **Parking** – Whether the restaurant offers parking

2.5 Restaurant Cuisines

- **Restaurant_ID** – Unique identifier for each restaurant
- **Cuisine** – Types of food served

3 ER Diagram

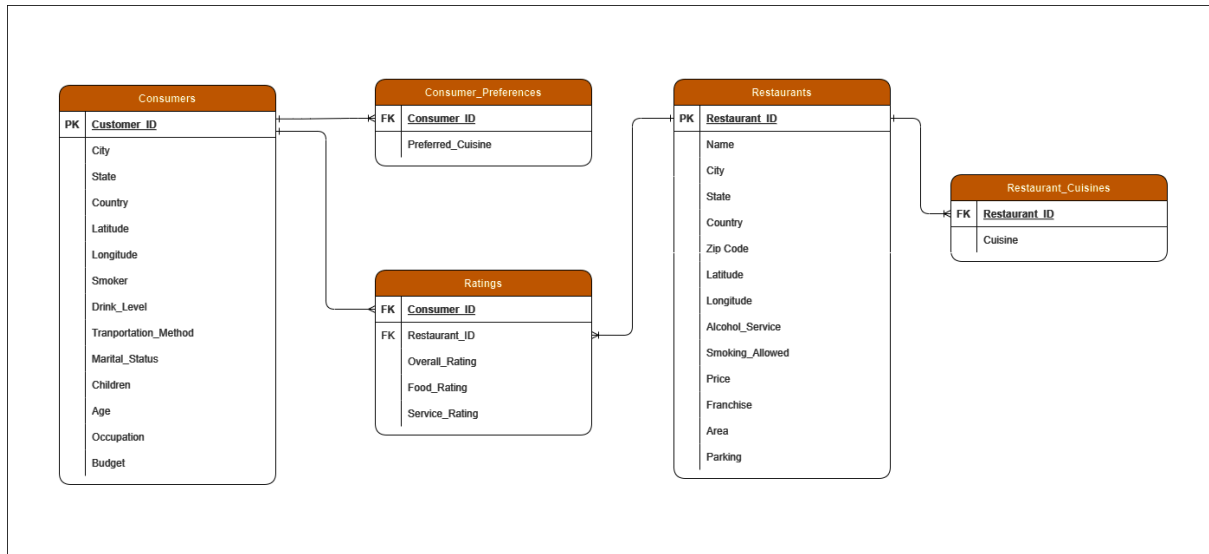


Figure 1: Entity Relationship Diagram

4 Data Cleaning

4.1 Steps to Import Data as a Folder

1. Get Data → More → All → Folder → Connect.
2. Select the dataset folder path and click *OK*.
3. Click *Transform Data*.
4. Duplicate the file.
5. Click on *Binary* to expand the dataset.
6. Repeat for all datasets.

4.2 Calculated Fields

4.2.1 Age Group

```
AgeGroup =
SWITCH(
  TRUE(),
  consumers[Age] <= 18, "Children and Adolescents",
  consumers[Age] <= 30, "Young Adults",
  consumers[Age] <= 45, "Adults",
  consumers[Age] <= 60, "Middle-aged Adults",
  "Seniors"
)
```

4.2.2 Service Rating Category

```
Service_Rating_Category = SWITCH(  
    TRUE(),  
    ratings[Service_Rating] = 0, "Unsatisfactory",  
    ratings[Service_Rating] = 1, "Satisfactory",  
    "Highly Satisfactory"  
)
```

4.2.3 Overall Rating Category

```
Overall_Rating_Category = SWITCH(  
    TRUE(),  
    ratings[Overall_Rating] = 0, "Unsatisfactory",  
    ratings[Overall_Rating] = 1, "Satisfactory",  
    "Highly Satisfactory"  
)
```

4.2.4 Food Rating Category

```
Food_Rating_Category = SWITCH(  
    TRUE(),  
    ratings[Food_Rating] = 0, "Unsatisfactory",  
    ratings[Food_Rating] = 1, "Satisfactory",  
    "Highly Satisfactory"  
)
```

5 Data Analysis

5.1 Local Insights

- Most consumers are from San Luis Potosí, followed by Cuernavaca, Morelos.
- Young adults under 30 form the majority across all states.
- Most consumers are non-smokers, with Jiutepec having a 100% non-smoking population.
- Most restaurants lack parking facilities.

5.2 Dining Insights

- High-priced restaurants are more likely to offer parking.
- San Luis Potosí has 84 restaurants, while Morelos and Tamaulipas each have 23.
- Non-franchise restaurants dominate the market.
- Mexican cuisine is the most preferred cuisine.

5.3 Hospitality Insights

- 66.92% of restaurants do not serve alcohol.
- 61% of consumers use public transportation.
- Restaurants offering wine and beer receive generally positive ratings.
- Approximately 73% of restaurants maintain smoke-free policies.

5.4 Behavior Insights

- Students constitute the majority of consumers.
- Drink preferences vary significantly across states.
- Most single consumers are non-smokers.
- Students predominantly have medium budgets.

5.5 Review Insights

5.5.1 Top Restaurants by Food Rating

- Tortas Locas Hipocampo
- Puesto de Tacos
- Cafeteria y Restaurante El Pacífico
- Gorditas Doña Gloria
- La Cantina Restaurante

5.5.2 Top Restaurants by Service Rating

- Tortas Locas Hipocampo
- Puesto de Tacos
- Cafeteria y Restaurante El Pacífico
- Gorditas Doña Gloria
- La Cantina Restaurante

5.5.3 Top Restaurants by Overall Rating

- Tortas Locas Hipocampo
- Puesto de Tacos
- Cafeteria y Restaurante El Pacífico
- La Cantina Restaurante
- Restaurant la Chalita